SPRiF: Spectral Projective Reset Integrate-and-Fire Neuron

Anonymous submission

Abstract

Temporal tasks often require information to persist across long intervals. Standard spiking neurons use the same membrane state to accumulate context, generate spikes, and receive post-spike reset. This coupling repeatedly perturbs the temporal trace that a neuron must preserve. We introduce **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), a spiking unit based on functional state decomposition. SPRiF separates a constrained spectral slow state, which provides real and rotational memory modes with no direct local reset update, from a fast discharge state that generates spikes and receives a learnable projective reset. The design preserves a resettable spike-generation pathway while insulating slow temporal memory from direct local post-spike correction. In descriptive cross-paper comparisons on five temporal classification benchmarks, SPRiF has the highest reported mean accuracy among the listed methods on three tasks and remains competitive on the other two. Among the tested variants, jointly merging the slow and fast states and reducing state dimensionality is associated with the largest observed performance differences; removing spectral rotation or fixing the reset direction produces smaller differences. A controlled spike-intervention task shows stable delay-period recall under reset stress, and impulse-response analysis reveals heterogeneous temporal kernels. These results support functional state decomposition as a practical neuron-level design principle for temporal SNNs.

Introduction

Spiking neural networks (SNNs) process temporal information through discrete spike events. Their temporal capacity depends on neuron dynamics, which govern integration, firing, and post-spike updates. When relevant information spans multiple timescales, these dynamics constrain the temporal patterns that an SNN can retain and exploit.

The leaky integrate-and-fire (LIF) neuron remains widely used because it is simple and compatible with surrogate-gradient training. Its extensions introduce adaptive thresholds (Baronig et al. 2025; Zhang et al. 2025b; Idrees et al. 2024), resonant dynamics (Higuchi et al. 2024; Huang et al. 2024; Zhang et al. 2025a), heterogeneous temporal decays (Wei et al. 2026; Deng et al. 2025; Fulleda-Garcia, Soldado-Magraner, and Margarit-Taulé 2026; Tiwari and Chauhan 2025), and modified reset (Deng et al. 2022; Huang et al. 2025; Li et al. 2025; Fang et al. 2023). Yet the spike-generating variable usually remains the reset target. A single membrane potential accumulates history, determines spike timing, and absorbs reset, so each spike perturbs its temporal trace.

Conventional LIF dynamics thus couple temporal integration, spike generation, and reset in one state; post-spike correction directly perturbs the retained input trace. The central issue is therefore not merely a lack of richer time constants, but the functional coupling of memory, spike generation, and reset. Separating memory from resettable discharge could preserve temporal context without sacrificing binary communication or enlarging network architecture.

We introduce **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), which implements functional state decomposition. Its slow spectral state combines one real exponential mode with a two-dimensional damped rotational mode to store temporal context without a direct local reset update. Its fast discharge state projects this context to a membrane readout, emits binary spikes, and receives a learnable projective reset along $[1, \lambda]^\top$. Trained with backpropagation through time and surrogate gradients, SPRiF confines direct local spike correction to the fast state while the slow state evolves continuously.

Unlike conventional multi-timescale models, whose parallel channels differ in decay but remain coupled to readout and reset, SPRiF separates state functions. Its slow state stores temporal context, whereas its fast state implements readout and reset; the reset operator has no direct update path to the slow state. The distinction is structural rather than a consequence of heterogeneous time constants alone.

We evaluate SPRiF on five benchmarks spanning sequential images, physiological signals, speech, and event-based digits. In descriptive cross-paper comparisons, SPRiF has the highest reported mean accuracy among the listed methods on S-MNIST, PS-MNIST, and QTDB, and remains competitive on GSC and SHD. Its reported end-to-end parameter count is lower than that of the highest-accuracy listed baseline on four datasets and comparable on QTDB. Ablations, controlled spike intervention, and impulse-response analyses examine rotation, state separation, and projective reset.

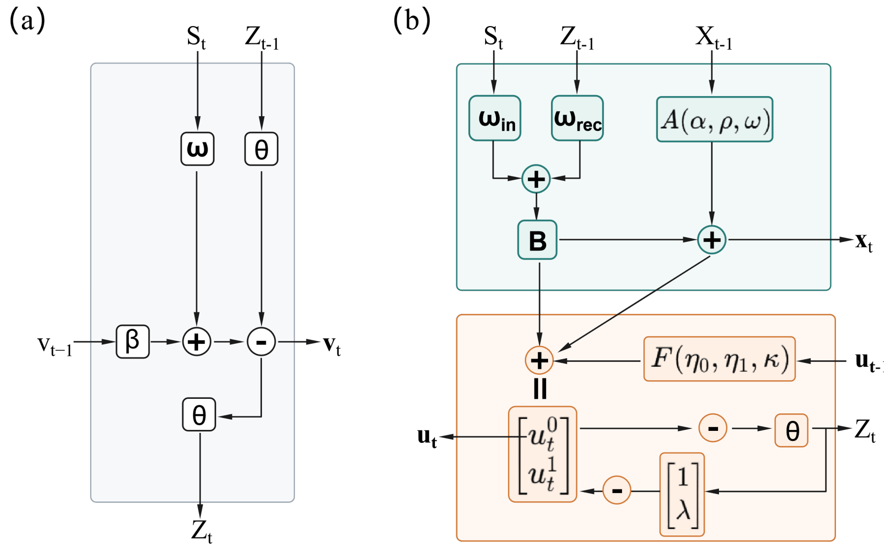

Figure 1: Architectures of (a) LIF and (b) SPRiF. SPRiF separates a slow memory state with no direct local reset update from a resettable fast discharge state.

The main contributions of our work are summarized below.

1. We formulate functional state decomposition, which separates temporal memory, spike readout, and post-spike reset. This principle is orthogonal to rate decomposition and single-state reset.
2. SPRiF instantiates this principle with a constrained spectral slow state that combines real exponential decay and damped rotational modes, and a fast discharge state with a learned projective reset that leaves the slow state free of direct local reset updates.
3. We evaluate SPRiF on five temporal benchmarks, three mechanism ablations, a controlled spike-intervention experiment, and analyses of learned temporal kernels. Code will be released upon acceptance.

Related Work

Neuron Dynamics for Temporal Processing

LIF extensions enrich temporal dynamics through adaptive variables, heterogeneous decays, and temporal gates (Yao et al. 2021; Baronig et al. 2025; Deng et al. 2025; Tiwari and Chauhan 2025). These approaches provide rate decomposition, in which parallel components evolve at different timescales but remain coupled to membrane readout and reset.

Resonate-and-fire variants use damped oscillations (Izhikevich 2001; Higuchi et al. 2024; Huang et al. 2024; Zhang et al. 2025a), whereas compartmental and heterogeneous models introduce dendritic structure and dynamics (Yin et al. 2024; Zheng et al. 2024; Stevenson and Kappel 2023; Lange et al. 2021). Multi-compartment neurons such as LSTM-LIF separate a dendritic long-term memory compartment from somatic discharge (Zhang et al. 2023), and recent modular SNNs explicitly decouple a memory module from a spiking module and its reset (Zhang 2025a). Structured state-space dynamics have also been incorporated into spiking neurons (Fabre 2025; Shen et al. 2025). Relatedly, DMP-SNN (Sun et al. 2026) combines a layer-shared, low-dimensional memory with a fast feed-forward spiking pathway for efficient long-horizon processing. SPRiF instead realizes neuron-level separation between a spectral slow state and a fast state with projective reset.

Reset and Functional State Separation

Existing reset mechanisms soften, adapt, or remove the post-spike update (Deng et al. 2022; Li et al. 2025; Huang et al. 2025; Zhang 2025b; Fang et al. 2023), but still modify the spike-generating state itself or remove reset from that state. SPRiF instead confines a learnable projective reset to the fast discharge state and introduces no direct local reset update to the slow spectral state, distinct from both scalar or adaptive reset (Deng et al. 2022; Li et al. 2025; Huang et al. 2025; Fang et al. 2023; Zhang 2025b) and module-level reset decoupling (Zhang 2025a). Network-level temporal architectures (Ding et al. 2025b, 2024, 2025a) are complementary to this neuron-level separation.

Method

Limitation of Conventional LIF Dynamics

For presynaptic activities $s_{i,t}$, a standard leaky integrate-and-fire (LIF) neuron receives input current

$$I_t = \sum_i w_i s_{i,t} + b, \tag{1}$$

where w_i is the synaptic weight of input i and b is a bias term. Its membrane update is

$$v_t = \beta v_{t-1} + I_t - z_{t-1}\theta, \tag{2}$$
$$z_t = H(v_t - \theta). \tag{3}$$

Here, $\beta \in (0,1)$ is the leak factor, H is the Heaviside step function, and θ is the firing threshold. The scalar v_t carries temporal context, determines spike timing, and receives the preceding reset. Unrolling over k steps gives

$$v_t = \beta^k v_{t-k} + \sum_{\ell=0}^{k-1} \beta^\ell \big(I_{t-\ell} - \theta z_{t-\ell-1}\big). \tag{4}$$

Input and reset histories are filtered through the same recurrence, so reset is inseparable from the state that transports temporal information. This motivates distinct memory and post-spike discharge states.

SPRiF Spiking Neuron Model

Figure 1 contrasts the two architectures. For layer input $\mathbf{s}_t$, the input current is

$$I_t = W_{\mathrm{in}}\mathbf{s}_t + W_{\mathrm{rec}}\mathbf{z}_{t-1} + b, \tag{5}$$

where the recurrent term is omitted in feed-forward layers. In the single-neuron equations below, I_t denotes the scalar current received by that neuron. Each neuron j maintains a three-dimensional slow state $\mathbf{x}_t \in \mathbb{R}^3$ and a two-dimensional fast state $\mathbf{u}_t \in \mathbb{R}^2$; $\widetilde{\mathbf{u}}_t$ denotes the pre-reset fast state. Its dynamics are

$$\begin{cases} \mathbf{x}_t = A\mathbf{x}_{t-1} + BI_t, \\ \widetilde{u}_t^0 = \eta_0 u_{t-1}^0 + \kappa u_{t-1}^1 + (G\mathbf{x}_t)^0, \\ \widetilde{u}_t^1 = \eta_1 u_{t-1}^1 + (G\mathbf{x}_t)^1, \\ v_t = \widetilde{u}_t^0, \qquad z_t = H(v_t - \theta), \\ \mathbf{u}_t = \widetilde{\mathbf{u}}_t - z_t\theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}. \end{cases} \tag{6}$$

The update follows $I_t \to \mathbf{x}_t \to \widetilde{\mathbf{u}}_t \to v_t \to z_t \to \mathbf{u}_t$. The slow state is updated before spike generation; the fast state is then read out, thresholded, and reset. The direct local reset update therefore acts only on the fast path, although $W_{\mathrm{rec}}\mathbf{z}_t$ lets spikes influence the slow state at later timesteps in recurrent layers.

Equation (6) is written for one neuron, with indices suppressed except in λ_j. Except for W_{in} and W_{rec}, the listed dynamics parameters are per-neuron. Layers apply the update at each timestep, and recurrence is optional.

Spectral Slow-Memory Dynamics

The slow-state update in Eq. (6) retains temporal context without direct spike-triggered correction. Rather than learning an unconstrained transition, SPRiF uses a three-dimensional real-valued system with one real decay mode and one complex-conjugate rotational pair.

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho\cos\omega & -\rho\sin\omega \\ 0 & \rho\sin\omega & \rho\cos\omega \end{bmatrix}, \quad B = \begin{bmatrix} 1-\alpha \\ 1-\rho \\ 0 \end{bmatrix}. \tag{7}$$

The spectrum of A is

$$\mathrm{eig}(A) = \left\{\alpha, \rho e^{\mathrm{i}\omega}, \rho e^{-\mathrm{i}\omega}\right\}. \tag{8}$$

The first coordinate is an exponential trace, $x_t^0 = \alpha x_{t-1}^0 + (1-\alpha)I_t$, while (x_t^1, x_t^2) form a damped rotation driven through the first oscillatory coordinate. We parameterize

$$\alpha = \sigma(\alpha_{\mathrm{raw}}), \quad \rho = \sigma(\rho_{\mathrm{raw}}), \quad \omega = \pi\sigma(\omega_{\mathrm{raw}}), \tag{9}$$

where σ is the logistic sigmoid. Thus $\alpha, \rho \in (0,1)$ and $\omega \in (0,\pi)$ place every eigenvalue of A strictly inside the unit disk. The parameters α, ρ, and ω control real decay, rotational damping, and rotational frequency, respectively, yielding a stable, interpretable spectrum.

The resulting temporal basis is explicit. For a unit impulse entering the slow state, the response after k subsequent transitions is

$$A^k B = \begin{bmatrix} (1-\alpha)\alpha^k \\ (1-\rho)\rho^k\cos(k\omega) \\ (1-\rho)\rho^k\sin(k\omega) \end{bmatrix}. \tag{10}$$

The real coordinate supplies a monotone exponential trace, whereas the rotational pair supplies phase-shifted, damped oscillations. Together they form a compact temporal basis with distinct decay and frequency controls. Because the slow state is computed before spikes and absent from the reset equation, its dynamics remain separate from post-spike correction.

Fast Discharge Dynamics and Projective Reset

The fast-state equations in Eq. (6) read out slow context and implement resettable discharge. The learned coefficients $\eta_0, \eta_1 \in (0,1)$ are fast leaks, κ couples the second fast coordinate to the first, and $G \in \mathbb{R}^{2\times 3}$ projects slow dynamics into discharge space.

The first fast coordinate is the membrane potential $v_t = \widetilde{u}_t^0$, and the forward pass generates $z_t = H(v_t - \theta)$ with a hard threshold. During optimization, only the threshold derivative is replaced by the multi-Gaussian surrogate (Yao et al. 2021); the forward dynamics remain unchanged.

After a spike, SPRiF applies the following projective reset only to the fast state.

$$\mathbf{u}_t = \widetilde{\mathbf{u}}_t - z_t\theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}, \tag{11}$$

where λ_j is learned independently for neuron j. Setting $\lambda_j = 0$ recovers scalar subtraction on the membrane coordinate; otherwise, reset modifies both fast coordinates along a learned direction. Thus, λ_j controls post-spike correction rather than the temporal memory transition and introduces no direct local update term for $\mathbf{x}_t$.

Reset-Memory Separation Analysis

The preceding equations distinguish the direct effect of reset from the ordinary temporal propagation of spike signals. Relative to the pre-reset states, the reset-induced increments are

$$\Delta\mathbf{x}_t^{\mathrm{reset}} = \mathbf{0}, \qquad \Delta\mathbf{u}_t^{\mathrm{reset}} = -z_t\theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}. \tag{12}$$

In LIF, the reset contribution $-z_{t-1}\theta$ is accumulated through the same recurrence as input history. In SPRiF, direct local

Table 1: Comparison across five temporal benchmarks. "Rec" denotes recurrent synaptic connections; DMP-SNN uses layer-shared state-space memory. Parameters are reported in millions (M), and SPRiF entries (bold) are mean±std over 5 seeds.

Datasets	Method	Network	Parameters (M)	Accuracy (%)
S-MNIST	GLIF$_{\text{NeurIPS, 2022}}$ (Fang et al. 2022)	Rec	0.15	96.64
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yao et al. 2021)	Rec	0.15	98.70
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.08	98.90
	BRF$_{\text{ICLR, 2024}}$ (Higuchi et al. 2024)	Rec	0.069	99.10
	TC-LIF$_{\text{AAAI, 2024}}$ (Yin et al. 2024)	Rec	0.15	99.20
	DMP-SNN (II)$_{\text{Nat. Mach. Intell., 2026}}$ (Sun et al. 2026)	DMP	0.073	99.20
	SPRiF (ours)	**Rec**	**0.067**	**99.28 ± 0.06**
PS-MNIST	GLIF$_{\text{NeurIPS, 2022}}$ (Fang et al. 2022)	Rec	0.15	90.47
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yao et al. 2021)	Rec	0.15	94.30
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.08	94.52
	BRF$_{\text{ICLR, 2024}}$ (Higuchi et al. 2024)	Rec	0.069	95.20
	TC-LIF$_{\text{AAAI, 2024}}$ (Yin et al. 2024)	Rec	0.15	95.36
	DMP-SNN (I)$_{\text{Nat. Mach. Intell., 2026}}$ (Sun et al. 2026)	DMP	0.061	95.50
	SPRiF (ours)	**Rec**	**0.067**	**95.86 ± 0.14**
QTDB	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yao et al. 2021)	Rec	0.0018	85.90
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.00178	86.35
	BRF$_{\text{ICLR, 2024}}$ (Higuchi et al. 2024)	Rec	0.00173	87.00
	SE-adLIF$_{\text{Nat. Commun., 2025}}$ (Baronig et al. 2025)	Rec	0.0018	86.88
	SPRiF (ours)	**Rec**	**0.00177**	**88.43 ± 0.36**
GSC	SNN-SFA$_{eLife,2021}$ (Salaj et al. 2021)	Rec	4.19	91.21
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yao et al. 2021)	Rec	0.31	92.10
	RadLIF$_{\text{Front. Neurosci., 2022}}$ (Kamata et al. 2022)	Rec	1.20	94.51
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.13	93.80
	TC-LIF$_{\text{AAAI, 2024}}$ (Yin et al. 2024)	Rec	0.20	94.84
	SPRiF (ours)	**Rec**	**0.13**	**94.55 ± 0.31**
SHD	Heterogeneous SNN$_{\text{Nat. Commun., 2021}}$ (Lange et al. 2021)	Rec	0.11	82.50
	ASRNN$_{\text{Nat. Mach. Intell., 2021}}$ (Yao et al. 2021)	Rec	0.14	90.40
	DH-SNN$_{\text{Nat. Commun., 2024}}$ (Zheng et al. 2024)	Rec	0.05	91.34
	BRF$_{\text{ICLR, 2024}}$ (Higuchi et al. 2024)	Rec	0.11	91.70
	TC-LIF$_{\text{AAAI, 2024}}$ (Yin et al. 2024)	Rec	0.14	88.91
	MPS-SNN$_{\text{ICLR, 2025}}$ (Ding et al. 2025b)	Rec	0.39	91.19
	DGN$_{\text{ICLR, 2026}}$ (Bai, Wang, and Yu 2026)	Rec	0.22	87.78
	SPRiF (ours)	**Rec**	**0.05**	**91.52 ± 0.46**

reset is confined to $\mathbf{u}_t$, leaving $\mathbf{x}_t$ unchanged at the same timestep.

The slow-state recurrence makes this distinction explicit. Iterating the first line of Eq. (6) yields

$$\mathbf{x}_t = A^t \mathbf{x}_0 + \sum_{\tau=1}^{t} A^{t-\tau} B I_\tau. \tag{13}$$

No direct local reset term appears in Eq. (13); holding I_t fixed gives $\partial \mathbf{x}_t / \partial z_t = \mathbf{0}$ for the reset operation. In recurrent layers, $\mathbf{z}_t$ affects $\mathbf{x}_{t+1}$ through $W_{\text{rec}}\mathbf{z}_t$. SPRiF therefore separates direct local post-spike reset from ordinary network communication.

Experiments

Benchmarks and Protocol

We evaluate SPRiF on five temporal classification benchmarks spanning sequential images (S-MNIST and PS-MNIST), physiological signals (QTDB), frame-based speech (GSC), and event-driven speech (SHD). We train all models for a fixed number of epochs. These tasks differ substantially in input encoding, sequence length, and temporal structure, allowing us to test whether the proposed neuron dynamics transfer beyond a single temporal modality.

We compare SPRiF with published recurrent SNNs that use gated, adaptive, dendritic, resonate-and-fire, and other temporal neuron dynamics. The architectures and optimization protocols are adapted to each benchmark, whereas the reported SPRiF variants retain the same neuron definition. Complete dataset preprocessing, task-specific architectures, optimization settings, initialization procedures, and reproducibility details are provided in the technical appendix. Classification accuracy is reported in Table 1.

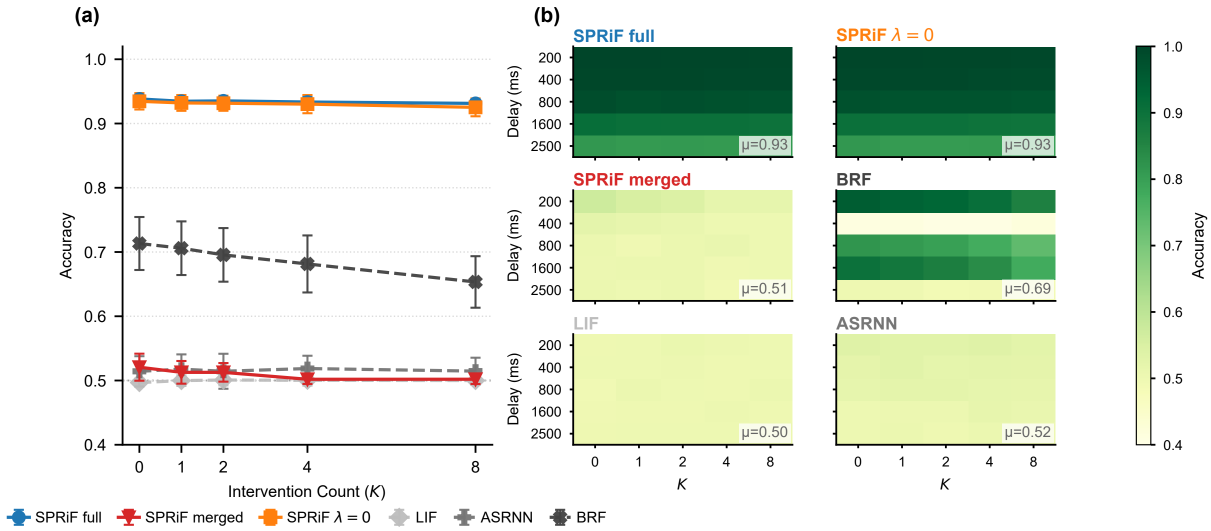

Figure 2: Controlled spike-intervention results (SI-DMS). (a) Accuracy versus intervention count K for the three SPRiF mechanism variants (solid, saturated) and three external baselines (dashed, gray); values are 3-seed means $\pm$ seed standard deviations. (b) Per-model delay-by-K accuracy grids (3-seed mean) for all six models with a shared colorbar; μ denotes the per-model mean across the grid.

Table 2: Mechanism ablations (accuracy, %).

Dataset	Full*	ω=0	Merged	λ=0
PS-MNIST	**95.86**	92.29	93.44	95.32
GSC	**94.55**	93.83	90.05	94.22
QTDB	**88.43**	87.78	83.39	87.40

*Full values are the 5-seed mean from Table 1.

Main Results

Table 1 summarizes the benchmark comparison. SPRiF obtains the highest reported accuracy among the listed methods on S-MNIST, PS-MNIST, and QTDB, while remaining competitive on GSC and SHD. It also has a lower reported parameter count than the highest-accuracy listed baseline on four datasets and a comparable count on QTDB. These results span five distinct temporal input regimes, showing consistent patterns across the tested settings. Because the published systems use task-specific architectures and training protocols, the table provides descriptive cross-paper context rather than a matched significance or efficiency comparison.

Mechanism Ablations

We isolated three design choices on PS-MNIST, GSC, and QTDB. Ablation A set $\omega = 0$, removing rotational coupling while retaining the real decay path. Ablation B merged memory and discharge into a single three-dimensional state, whose first coordinate served as both membrane and reset target. Ablation C fixed $\lambda = 0$, reducing Eq. (11) to scalar reset. All variants otherwise retained the task-specific training configuration.

The merged variant jointly collapses the two state roles and reduces total state dimensionality, so this comparison does not isolate separation alone. With this qualification, Table 2 shows a consistent ordering across all three datasets. The merged variant has the largest observed accuracy decrease, 2.42–5.04 points. Removing rotation is associated with a 0.65–3.57-point decrease, with the largest difference on the long permuted sequence, while fixing the reset direction is associated with a smaller but consistent 0.33–1.03-point decrease. Across the tested settings, the joint separation-and-capacity change is associated with the largest performance difference, with spectral rotation and a learnable reset direction providing further refinement.

Controlled Spike Intervention

To test whether slow–fast separation protects memory under reset stress, we designed a Spike-Intervention Delayed Match-to-Sample (SI-DMS) task. A cue was followed by a delay, during which selected hidden neurons were induced to spike at controlled times. A match/non-match probe then tested recall. At each intervention timestep, the membrane coordinates of 10% of hidden neurons were placed above threshold before spike computation, so the standard reset mechanism was applied. We evaluated $K \in \{0, 1, 2, 4, 8\}$ intervention times. The maximum-stress setting, $K = 8$, was not used during training. If reset reaches the memory-bearing state, these interventions should impair delay-period recall. If the slow state is structurally insulated, recall should remain more stable.

Table 3 reports clean ($K = 0$) and maximum-stress ($K = 8$) accuracy averaged over three seeds. At maximum stress, the two separated variants maintained high accuracy with only small losses. SPRiF full reached 92.8% clean accuracy and lost 0.7 percentage points, while SPRiF λ=0 showed

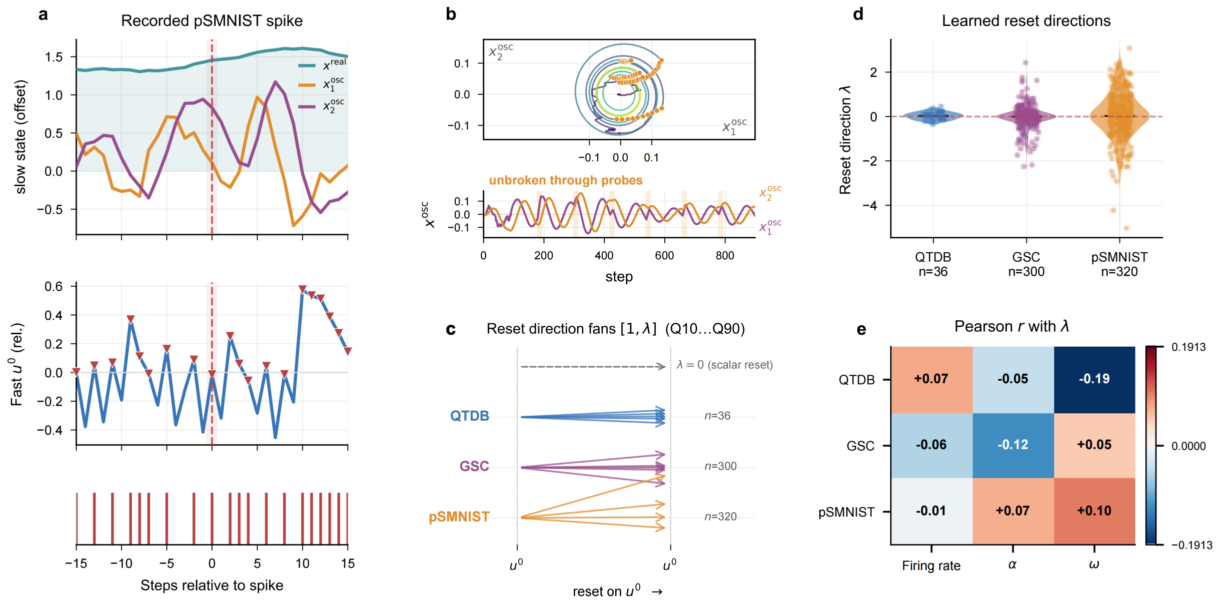

Figure 3: Functional decomposition and learned projective reset. (a) Slow and fast states around a recorded PS-MNIST spike. (b) Oscillatory slow-state trajectories across cue–delay probe windows. (c) Representative reset directions $[1, \lambda]$. (d) Task-wise distributions of λ. (e) Pearson correlations between λ and firing rate, α, and ω.

Table 3: SI-DMS results (3-seed average; accuracy in %; drop in percentage points). Clean and maximum-stress accuracies correspond to $K = 0$ and $K = 8$, respectively. External baselines are marked (ext.). The two separated SPRiF variants maintain high accuracy under intervention.

Model	Clean	Stress	Drop
SPRiF full	93.5	92.9	0.6
SPRiF λ=0	92.8	92.1	0.7
SPRiF merged	51.7	50.4	1.3
LIF (ext.)	50.0	50.2	−0.2
ASRNN (ext.)	52.0	51.7	0.3
BRF (ext.)	70.4	64.9	5.5

a comparable 0.6-point drop. This pattern is consistent with preserved delay-period recall and indicates that structural separation, rather than λ alone, was important in the tested setting. BRF decreased from 70.4% to 64.9%. The merged variant remained near chance (51.7% clean and 50.4% under maximum stress), as did LIF and ASRNN, so their clean-to-stress differences are interpreted only within that performance range. Figure 2 shows the delay-by-intervention grid, with complete grids and integrity audits provided in the appendix.

Dynamical Analysis

The update equations ensure that direct local reset does not modify $\mathbf{x}_t$, and SI-DMS tests the functional consequence of this design under controlled interventions. Figure 3a shows the resulting state dynamics around recorded spikes, while Fig. 3b shows continuous slow-state oscillation across probe windows in a cue–delay task. These traces illustrate continuous slow-state evolution but are not, by themselves, an event-level reset test. Figure 3c–e examines the learned projective reset directions.

Projective reset did not collapse to a shared direction. Across QTDB, GSC, and PS-MNIST, the learned λ values spanned both signs (Fig. 3c–d). Their correlations with firing rate, α, and ω were weak, with $|r| \leq 0.191$ (Fig. 3e). Together with the $\lambda = 0$ ablation difference, these observations support diverse learned reset directions in the fast state. The analysis characterizes reset geometry in the tested networks; biological interpretation lies outside its scope.

To characterize learned memory kernels, we injected a unit impulse into trained neurons and recorded 100 steps of intrinsic slow-state evolution. With no subsequent input, this diagnostic isolates the temporal basis $A^k B$ in Eq. (10) from task-specific input activity and subsequent post-spike correction. Unlike SI-DMS, which probes recall under externally induced spikes, the impulse response exposes the intrinsic slow transition. We sampled neurons across α quantiles and aggregated real-coordinate spectra across all neurons, providing complementary trace-level and task-wide views in Fig. 4.

The α-stratified real coordinates span rapid and persistent responses, while the rotational coordinates vary in phase and frequency. At the population level, the median normalized real-coordinate spectra concentrate at low frequencies, and their 10–90% ranges show substantial within-task variation. Normalization focuses this comparison on response shape rather than absolute amplitude. These observations connect

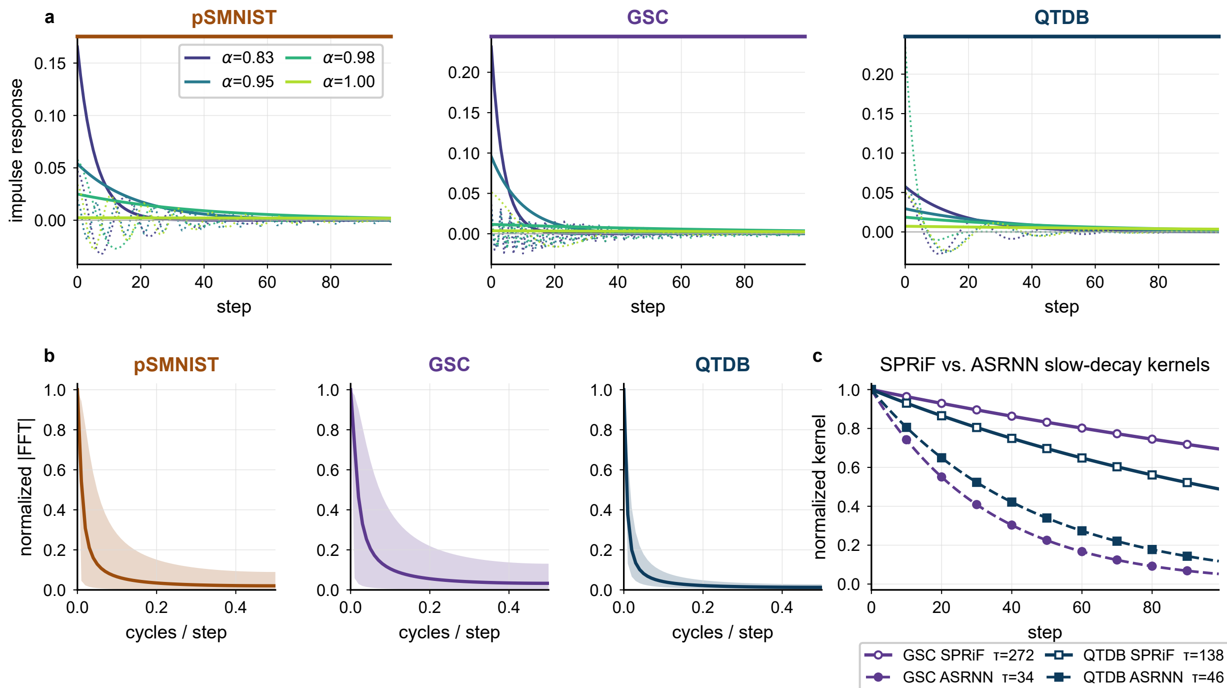

Figure 4: Learned temporal kernels. (a) Real (solid) and oscillatory (dotted) coordinates of four α-quantile SPRiF neurons per task over 100 post-impulse timesteps. (b) Median normalized real-coordinate spectrum with the 10–90% range. (c) Representative SPRiF real-coordinate and available trained ASRNN membrane kernels on GSC and QTDB, with derived time constants τ.

the learned kernel diversity to the constrained parameterization. The parameter α controls monotone retention, while ρ and ω govern rotational damping and phase evolution.

On GSC and QTDB, selected slow-decay SPRiF real modes have derived time constants of approximately 272 and 138 steps, respectively, compared with 34 and 46 steps for the available trained ASRNN membrane kernels. This checkpoint-availability-limited diagnostic characterizes the observed responses on the two tasks rather than establishing a general cross-model advantage. Additional impulse-response and spectral analyses are provided in the appendix.

Conclusion

SPRiF is built on the principle that the state storing temporal context need not receive a direct local reset update after a spike. A constrained spectral slow state supplies real and rotational memory modes, while a fast discharge state produces binary spikes and receives a learnable projective reset. In descriptive cross-paper comparisons across five temporal benchmarks, SPRiF has the highest reported mean accuracy among the listed methods on three tasks and remains competitive on the other two, with reported parameter counts that are comparable to or lower than those of the highest-accuracy listed baselines. Among the tested ablations, the variant that jointly collapses the slow and fast states and reduces state dimensionality has the largest observed degradation, while removing rotation or fixing the reset direction yields smaller decreases. Controlled spike-intervention and temporal-kernel analyses provide complementary mechanistic evidence for the proposed design. These findings support functional state decomposition as a reusable neuron-level design principle. Larger-scale evaluation and neuromorphic-hardware deployment remain open.

References

Bai, Q.; Wang, H.; and Yu, Q. 2026. DGN: A Brain-Inspired Gating Mechanism Unlocks Robust Computation in Spiking Neural Networks. In *International Conference on Learning Representations (ICLR)*.

Baronig, M.; et al. 2025. Advancing Spatio-Temporal Processing through Adaptation in Spiking Neural Networks. *Nature Communications*, 16.

Deng, S.; et al. 2022. InfLoR-SNN: Reducing Information Loss for Spiking Neural Networks. In *European Conference on Computer Vision (ECCV)*.

Deng, Z.; Liu, Z.; Wang, J.; Chen, S.; Wei, X.; and Yu, Q. 2025. HetSyn: Versatile Timescale Integration in Spiking Neural Networks via Heterogeneous Synapses. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Ding, J.; Yu, Z.; Huang, T.; and Liu, J. K. 2024. Enhancing the Robustness of Spiking Neural Networks with Stochastic Gating Mechanisms. In *AAAI Conference on Artificial Intelligence*, volume 38, 492–502.

Ding, J.; Yu, Z.; Liu, J. K.; and Huang, T. 2025a. Neuromorphic Computing Paradigms Enhance Robustness through Spiking Neural Networks. *Nature Communications*, 16: 10175.

Ding, Y.; Zuo, L.; Jing, M.; He, P.; and Deng, H. 2025b. Rethinking Spiking Neural Networks from an Ensemble Learning Perspective. In *International Conference on Learning Representations (ICLR)*.

Fabre, O. 2025. SiLIF: Structured State Space Model Dynamics and Parametrization for Spiking Neurons. ArXiv:2506.06374.

Fang, W.; Chen, Y.; Ding, J.; Zhai, Z.; Yu, X.; Zhang, T.; and Huang, T. 2022. GLIF: A Unified Gated Leaky Integrate-and-Fire Neuron for Spiking Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Fang, W.; et al. 2023. Parallel Spiking Neurons with High Efficiency and Ability to Learn Long-term Dependencies. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Fulleda-Garcia, A.; Soldado-Magraner, S.; and Margarit-Taulé, J. M. 2026. Multi-Timescale Conductance Spiking Networks with Rich Firing Dynamics for Enhanced Temporal Processing. *arXiv preprint*. ArXiv:2605.11835.

Higuchi, K.; et al. 2024. Balanced Resonate-and-Fire Neurons. In *International Conference on Machine Learning (ICML)*.

Huang, Y.; Liu, Z.; Feng, C.; Lin, X.; Ren, H.; Fu, H.; Zhou, Y.; Xing, H.; and Cheng, B. 2024. PRF: Parallel Resonate and Fire Neuron for Long Sequence Learning in SNNs. ArXiv:2410.03530.

Huang, Z.; Meng, W.; Liu, Q.; Chen, K.; and Ma, L. 2025. AR-LIF: Adaptive Reset Leaky Integrate-and-Fire Neuron for Spiking Neural Networks. ArXiv:2507.20746.

Idrees, S.; Manookin, M. B.; Rieke, F.; Field, G. D.; and Zylberberg, J. 2024. Biophysical Neural Adaptation Mechanisms Enable Artificial Neural Networks to Capture Dynamic Retinal Computation. *Nature Communications*, 15: 5957.

Izhikevich, E. M. 2001. Resonate-and-Fire Neurons. *Neural Networks*, 14(6-7): 883–894.

Kamata, K.; Mukawa, Y.; Matsuda, S.; et al. 2022. A Surrogate Gradient Spiking Baseline for Speech Command Recognition. *Frontiers in Neuroscience*, 16.

Lange, M.; et al. 2021. Neural Heterogeneity Promotes Robust Learning. *Nature Communications*, 12.

Li, Y.; Zeng, X.; Xue, Z.; Zeng, P.; Zhang, Z.; and Wang, Y. 2025. Incorporating the Refractory Period into SNNs through Spike-Triggered Threshold Dynamics. In *ACM International Conference on Multimedia*.

Salaj, D.; Subramoney, A.; Kraisnikovic, C.; Bellec, G.; Legenstein, R.; and Maass, W. 2021. Spike Frequency Adaptation Supports Network Computations on Temporally Dispersed Information. *eLife*, 10: e65459.

Shen, G.; et al. 2025. Learning Long Sequences with Sparse and Parallel Spiking State Space Models. In *AAAI Conference on Artificial Intelligence*.

Stevenson, J.; and Kappel, D. 2023. Introducing the Dendrify Framework for Incorporating Dendrites to Spiking Neural Networks. *Nature Communications*, 14.

Sun, P.; Su, Z.; Achterberg, J.; Indiveri, G.; Goodman, D. F. M.; and Akarca, D. 2026. Algorithm–Hardware Co-Design of Neuromorphic Networks with Dual Memory Pathways. *Nature Machine Intelligence*, 8: 901–912.

Tiwari, A.; and Chauhan, A. 2025. Local Timescale Gates for Timescale-Robust Continual Spiking Neural Networks. ArXiv:2510.12843.

Wei, X.; Zhou, X.; Liu, X.; Zang, Y.; and Yu, Q. 2026. Recurrent Spiking Neural Networks with Bimodal Neuronal Time Scales for Learning Performance Enhancement. *Neural Networks*, 200: 108830.

Yao, Z.; Zeng, R.; Peng, Q.; Dong, J.; Xiao, Z.; and Xu, G. 2021. Accurate and Efficient Time-Domain Classification with Adaptive Spiking Recurrent Neural Networks. *Nature Machine Intelligence*, 3: 912–922.

Yin, B.; et al. 2024. TC-LIF: A Two-Compartment Spiking Neuron Model for Long-Term Sequential Modelling. In *AAAI Conference on Artificial Intelligence*.

Zhang, D.; Zhang, M.; Wang, S.; Wang, J.; Wei, W.; Ma, Z.; Wang, G.; Yang, Y.; and Li, H. 2025a. Dendritic Resonate-and-Fire Neuron for Effective and Efficient Long Sequence Modeling. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Zhang, E. 2025a. Revisiting Reset Mechanisms in Spiking Neural Networks for Sequential Modeling: Specialized Discretization for Binary Activated RNN. arXiv:2504.17751.

Zhang, E. 2025b. Revisiting Reset Mechanisms in Spiking Neural Networks: Specialized Discretization for Binary Activated RNN. ArXiv:2504.17751.

Zhang, S.; Qu, Y.; Ma, C.; Wu, J.; Li, H.; and Tan, K. C. 2023. Long Short-term Memory with Two-Compartment Spiking Neuron. arXiv:2307.07231.

Zhang, T.; Yu, K.; Zhang, J.; and Wang, H. 2025b. DA-LIF: Dual Adaptive Leaky Integrate-and-Fire Model for Deep Spiking Neural Networks. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1–5.

Zheng, H.; Zheng, Z.; Hu, R.; Xiao, B.; Wu, Y.; Yu, F.; Liu, X.; Li, G.; and Deng, L. 2024. Temporal Dendritic Heterogeneity Incorporated with Spiking Neural Networks for Learning Multi-Timescale Dynamics. *Nature Communications*, 15: 277.